OPSINTEL AI

# Evidence-based pipeline investigation

From failed overnight run to traceable next action - powered by Exasol Personal

EXATHON INDIA 2026 / TRACK 04: THE WORK BEHIND THE WORK

Public prototype • MIT licensed • Synthetic data

THE PROBLEM

# At 3:07 a.m., the alert says failed. The evidence is scattered.

MANUAL

## Evidence lives in separate views

This run expected 100 rows but received 80. The engineer still searches separate run, count, and log views.

OPSINTEL

01 Select the failed run

02 Review cited status, counts, and logs

03 Keep facts separate from candidate explanations

04 System abstains until a human verifies

**54% of survey respondents reported a significant outage costing more than $100k (Uptime Institute, 2024). OpsIntel time saving remains unmeasured.**

PRODUCT WORKFLOW

# Operational evidence flows through Exasol, FastAPI, and the dashboard

01 **Pipeline runs and logs** →

02 **Exasol SQL analytics** →

03 **FastAPI rules and candidates** →

04 **Dashboard + optional LLM narrative** →

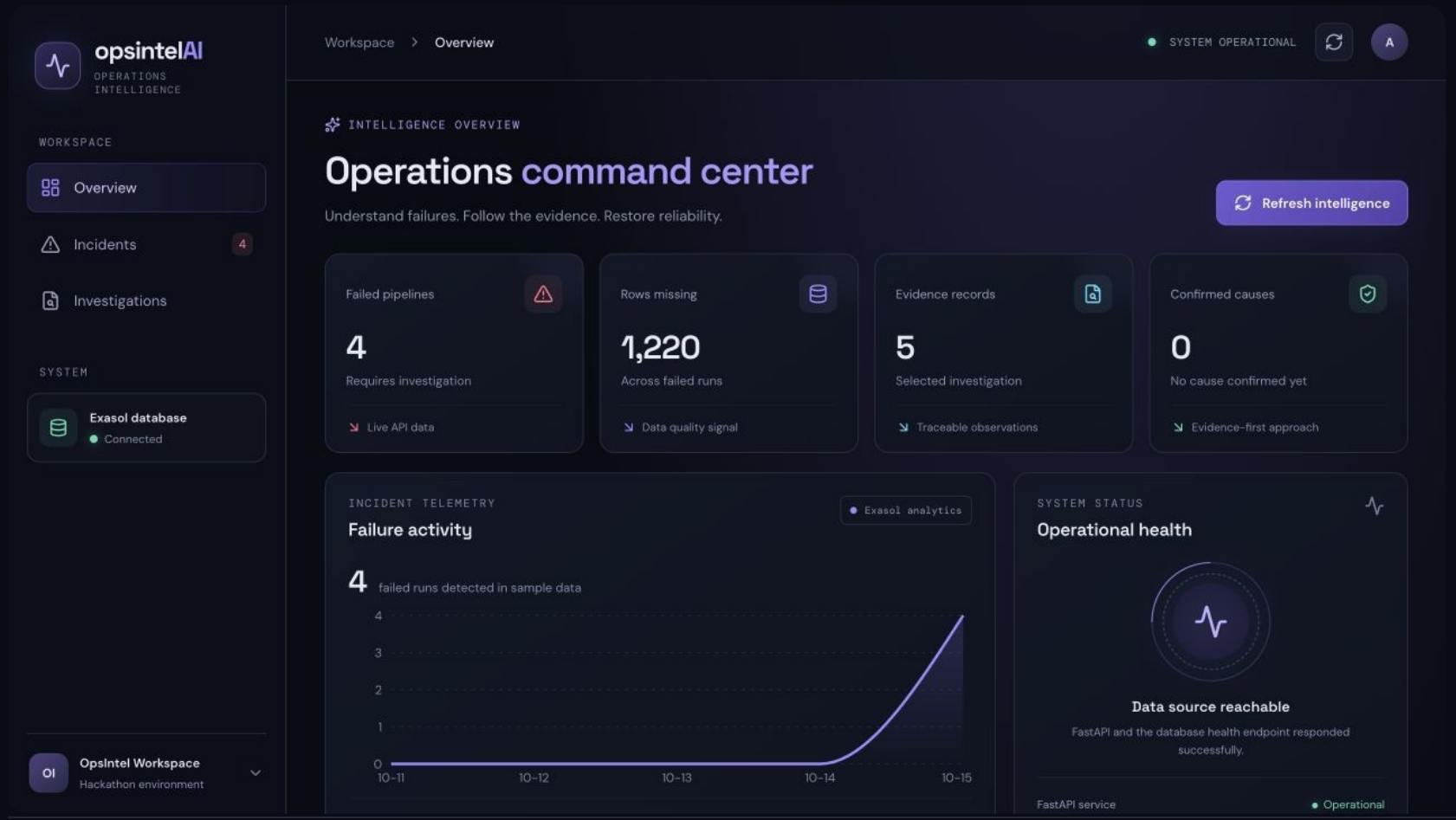
05 **Engineer verifies and confirms**

**Safety boundary**

Rules produce the category and candidates. The optional LLM only summarizes retrieved evidence. Confirmation and repairs stay human-controlled.

LIVE PRODUCT

# The dashboard reads incident telemetry calculated in Exasol




**4**

failed pipeline runs

**1,220**

aggregate missing rows in the captured 4-case baseline

✓ Captured baseline: 4 incidents
✓ Current seed: 11 incident scenarios
✓ All values are synthetic

INVESTIGATION EXPERIENCE

# Every recommendation points back to source evidence

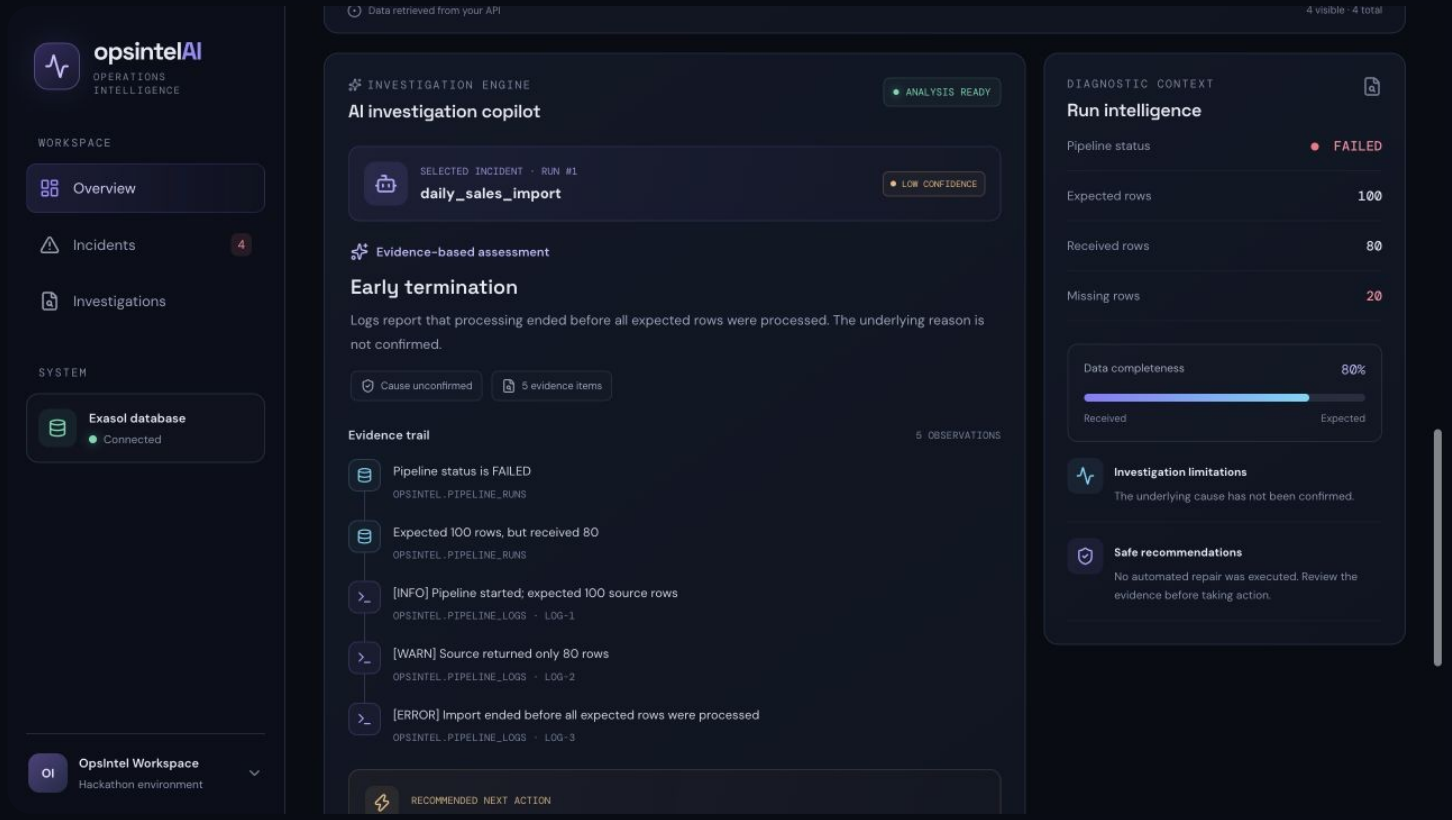
Selected failure vs last success

## daily_sales_import

FAILED VS LAST SUCCESS

Failed: 80 / 100 rows (80%)
Last success: 100 / 100 (100%)
Change: -20 rows / -20 pp

EVIDENCE TRAIL
PIPELINE_RUNS: status and row counts
LOG-1: expected 100 source rows
LOG-2: source returned only 80 rows
LOG-3: import ended before completion




CANDIDATES

Source shortfall or early termination
Confidence: low

HUMAN CONFIRMATION

Not yet verified
No automated repair

WHY EXASOL

# Exasol computes incident context before FastAPI responds

## ANALYTICAL FEATURES

01 Previous run status with LAG

02 Rolling actual-row baseline

03 Rolling failure rate

04 Missing rows and completeness

05 Parallel in-memory SQL execution

**LIVE QUERY USED BY /analytics/trend**

```
SELECT RUN_DAY, COUNT(*) AS TOTAL_RUNS,
       SUM(IS_INCIDENT) AS INCIDENT_COUNT,
       ROUND(AVG(COMPLETENESS_PCT), 1) AS AVG_COMPLETE
FROM OPSINTEL.PIPELINE_HEALTH_FEATURES
GROUP BY RUN_DAY;
```

**VERIFIED EXASOL OUTPUT**

**12 runs, 11 incidents, 64.1% average completeness**

The API reads this view directly. Verified against Exasol Personal.

TEMPORAL DEPENDENCY CORRELATION

# One upstream failure correlates with three downstream incidents

01

## Window analytics at scale

LAG, rolling baselines and failure rates compare each run with recent history inside Exasol.

02

## Failure cascade

CONNECT BY retains downstream failures within a 120-minute window. Temporal correlation only. Causation not confirmed.

03

## Operational evidence layer

FastAPI returns the same SQL-derived metrics and source records that judges can inspect in the dashboard.

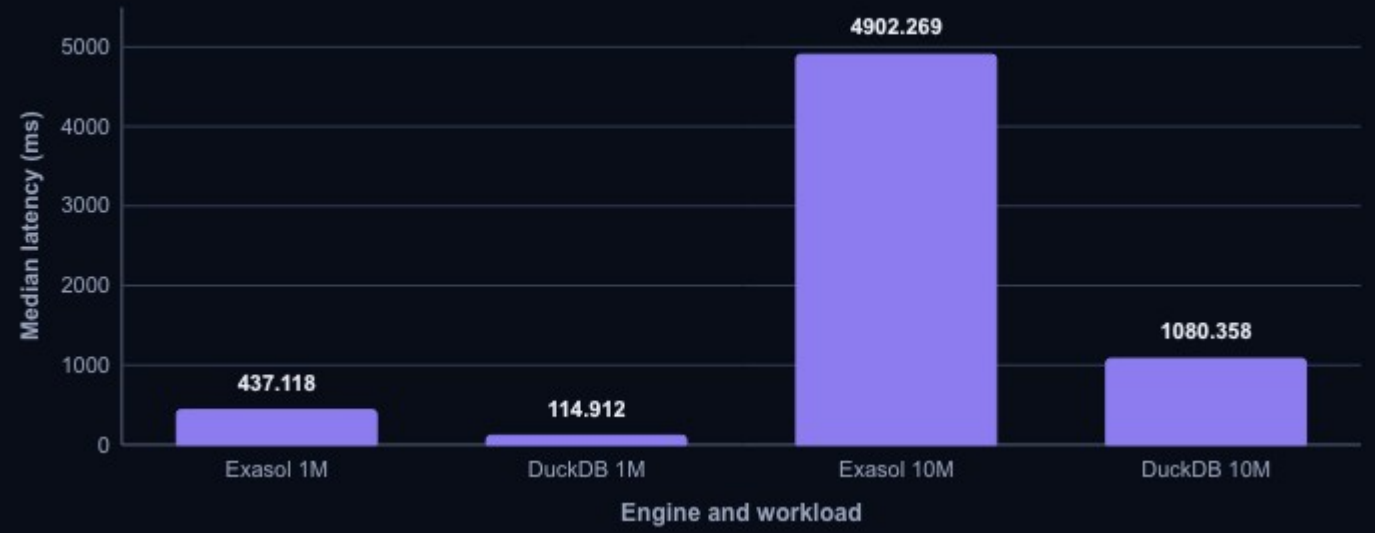
IMPLEMENTED CASCADE QUERY

**customer_sync** → **daily_reporting** · **customer_orders_sync** · **billing_customer_export**

Live result: customer_sync correlates with 3 later failed runs. Bulk import loads benchmark history; no UDF is implemented.

LOCAL SINGLE-USER BENCHMARK

# At 10M rows, DuckDB was 4.5× faster




Source figure (non-editable). 10M medians: Exasol 4902.269 ms; DuckDB 1080.358 ms.

**137 / 137**

automated tests passed

**6 / 9**

licensed public corpus
project holdout: 27 / 27

QUERY CACHE

**OFF**

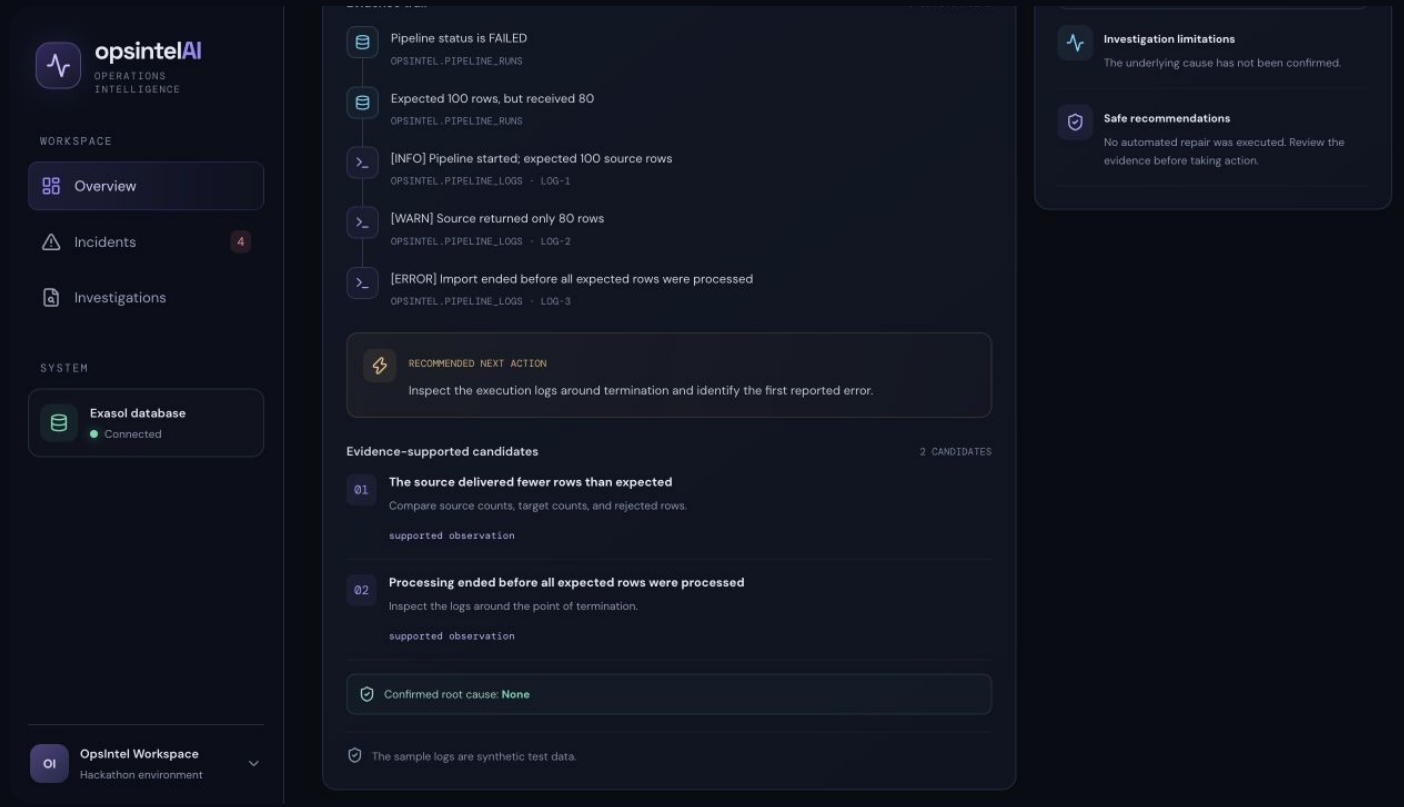
REPETITIONS

**7 RUNS**

Same deterministic query, 7 runs per engine, Exasol query cache OFF. 1M medians: Exasol 437 ms, DuckDB 115 ms. No production or concurrency claim.

TRUST AND SAFETY

# The product is designed to abstain before it overclaims




OBSERVATION

## Source-backed facts

CANDIDATE

## Evidence-supported possibility

CONFIRMATION

## Reserved for human verification

INSUFFICIENT EVIDENCE EXAMPLE
Worker exited with code 71. No diagnostic details. Result: unknown. The system says what to check next and cannot confirm a cause or run a repair.

VALIDATION ROADMAP

# Evidence status and next validation steps

## PROVEN NOW

01 Live Exasol-backed demo

02 Controlled 1M/10M comparison

03 137 automated tests

04 Clean clone: setup and checks passed

## DEFINED, NOT COMPLETED

01 External holdout: 0 reviewer cases

02 Independent LLM quality evaluation

03 Impact study: 0/40 trials (0/4 people)

04 Exasol workflow in a clean environment

## PRODUCTION WORK

01 Live Airflow / dbt connectors

02 Authentication and authorization

03 Concurrent workload testing

04 Audited retention and monitoring

**Clean-clone checks passed. Exasol workflow, external review and participant trials remain pending.**

OPSINTEL AI

# Follow the evidence. Investigate faster. Confirm carefully.

github.com/Anafa-thabassum/opsintel-ai

Live demo: observed facts, candidate causes, human verification